

Learning Seminar on D-modules 2024

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1 Introduction

Let X be a smooth, complex variety in what follows, $\mathcal{O}_X$ will denote the sheaf of regular functions on X .

Definition 1.1. We denote by Θ_X the sheaf of vector fields/derivations on X , i.e.

$$\begin{aligned}\Theta_X &= \text{Der}_{\mathbb{C}_X}(\mathcal{O}_X) \\ &:= \{\theta \in \text{End}_{\mathbb{C}_X}(\mathcal{O}_X) \mid \theta(fg) = \theta(f)g + f\theta(g), \forall f, g \in \mathcal{O}_X\}\end{aligned}$$

Remark 1.2. 1. X smooth $\implies \Theta_X$ locally free of rank $n = \dim(X)$.
2. Both Θ_X and $\mathcal{O}_X$ are subsheaves of $\text{End}_{\mathbb{C}_X}(\mathcal{O}_X)$.

Definition 1.3 (Main Definition). The sheaf of differential operators on X is

$$D_X := \langle \Theta_X, \mathcal{O}_X \rangle \subseteq \text{End}_{\mathbb{C}_X}(\mathcal{O}_X)$$

as a sub- $\mathbb{C}$ -algebra of $\text{End}(\mathcal{O}_X)$ generated by Θ_X and $\mathcal{O}_X$.

To better understand what this sheaf looks like, we describe it locally. To this end, let U be an affine open subset of X , and let $\{x_i, \partial_i\}$ be a local coordinate system $[x_i \in \mathcal{O}_X(U), \partial_i \in \Theta_X(U)]$ such that

$$\Theta_U = \bigoplus_{i=1}^n \mathcal{O}_U \partial_i$$

satisfying the following relations

$$[x_i, x_j] = 0, \quad [\partial_i, \partial_j] = 0, \quad [\partial_i, x_j] = \delta_{ij}$$

Then,

$$D_U := D_X|_U = \bigoplus_{\alpha \in \mathbb{N}^n} \mathcal{O}_U \partial^\alpha + x^\alpha$$

where $\mathbb{N}$ is the set of non-negative integers and $\partial^\alpha = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n}$.

Example 1.4. Weyl algebra. Fix positive n . The n -th Weyl algebra, D_n , over $\mathbb{C}$ is the ring of differential operators with complex polynomial coefficients in n -variables, i.e.

$$D_n = \mathbb{C}\{x_1, \dots, x_n, \partial_1, \dots, \partial_n\}$$

subject to $[x_i, x_j] = 0, \quad [\partial_i, \partial_j] = 0, \quad [\partial_i, x_j] = \delta_{ij}$ (Leibniz rule). This is a non-commutative $\mathbb{C}$ -algebra, but there is a sense in which D_n is "almost" commutative.

Example 1.5 (HTT, pg. 16). Let $\{x_i, \partial_i\}_{1 \leq i \leq n}$ be a local coordinate system on an affine open subset $U \subseteq X$. For

$$P = \sum_{\alpha \in \mathbb{N}^n} a_\alpha(x) \partial_x^\alpha \in D_X(U)$$

we define its total symbol $\sigma(P)(x, \xi)$ by

$$\sigma(P)(x, \xi) := \sum_{\alpha \in \mathbb{N}^n} a_\alpha(x) \xi^\alpha$$

For $P, Q \in D_X(U)$, let $R = PQ \in D_X(U)$. Show that

$$\sigma(R)(x, \xi) = \sum_{\alpha \in \mathbb{N}^n} \frac{1}{\alpha!} \partial_\xi^\alpha \sigma(P)(x, \xi) \cdot \partial_x^\alpha \sigma(Q)(x, \xi)$$

where $\alpha! = \alpha_1! \alpha_2! \dots \alpha_n!$ for each $\alpha \in \mathbb{N}^n$.

Solution. We will leave the general case as homework, but let's try out a special case for elucidation. Let $X = \mathbb{A}_{\mathbb{C}}^2$ and let $P = x_1 x_2 \partial_1$, $Q = x_1 \partial_2$, so that $R = PQ = x_1 x_2 \partial_2 + x_1^2 x_2 \partial_1 \partial_2$. Then

$$\begin{aligned} \sigma(P)(x, \xi) &= x_1 x_2 \xi_1, & \sigma(Q)(x, \xi) &= x_1 \xi_2 \\ \sigma(R)(x, \xi) &= x_1 x_2 \xi_1 + x_1^2 x_2 \xi_1 \xi_2 \end{aligned}$$

We verify the identity:

$$\begin{aligned} & \sum_{\alpha \in \mathbb{N}^n} \frac{1}{\alpha!} \partial_\xi^\alpha \sigma(P)(x, \xi) \cdot \partial_x^\alpha \sigma(Q)(x, \xi) \\ &= \frac{1}{0!0!} \sigma(P) \sigma(Q) + \frac{1}{1!0!} \partial_{\xi_1} \sigma(P) \cdot \partial_{x_1} \sigma(Q) \\ &= x_1^2 x_2 \xi_1 \xi_2 + x_1 x_2 \xi_2 = \sigma(R)(x, \xi). \end{aligned}$$

Definition 1.6 (Filtration). We define the order filtration $F_\bullet$ on D_U by:

$$F_l D_U := \sum_{|\alpha| \leq l} \mathcal{O}_U \partial_x^\alpha \quad \left(l \in \mathbb{N}, |\alpha| = \sum_i \alpha_i \right)$$

Remark 1.7. I. We can "globalize" this notion by gluing, i.e.

$$F_l D_X(V) = \{P \in D_X(V) \mid \text{res}_U^V P \in F_l D_X(U) \text{ } (\forall U)\}$$

or by

II. Set $F_0 D_X = \mathcal{O}_X$, $F_l D_X = \{P \in \text{End}_{\mathbb{C}_X}(\mathcal{O}_X) \mid [P, g] \in F_{l-1} D_X \text{ } (\forall g \in \mathcal{O}_X)\}$

Definition 1.8. The graded sheaf associated with the sheaf of filtered rings $(D_x, F_\bullet)$ is

$$gr^F D_X := \bigoplus_{l \geq 0} gr_l^F(D_X)$$

where $gr_l^F D_X := F_l D_X / F_{l-1} D_X$ and defining $F_{-1} D_X = 0$.

Remark 1.9. $gr^F D_X$ is a sheaf of commutative algebraic finitely generated over $\mathcal{O}_X$. Locally, we have

$$gr_l^F D_U = (F_l / F_{l-1}) D_U = \bigoplus_{|\alpha|=l} \mathcal{O}_U \xi^\alpha$$

and

$$gr^F D_U \cong \mathcal{O}_U[\xi_1, \dots, \xi_n]$$

In this way, we may think of D_X as "almost" commutative.

Definition 1.10. For a differential operator $P \in F_l D_U$, the section $\sigma_l(P) \in gr_l^F(D_U)$ is called the principal symbol of P .

We can give a more geometric and global identification of the graded ring $gr D_X$ with the ring of polynomial functions on the cotangent bundle as follows:

$$gr D_X \cong \pi_* \mathcal{O}_{T^*X} \cong Sym \Theta_X$$

We do this by locally regarding $\xi_1, \dots, \xi_n$ as the coordinate system of the cotangent space $\bigoplus_{i=1}^n \mathbb{C} dx_i$. Thus,

$$\begin{array}{ccc} \mathcal{O}_U[\xi_1, \dots, \xi_n] & \xrightarrow{\cong} & \pi_* \mathcal{O}_{T^*X}|_U \\ \parallel & & \\ gr D_X|_U & & \end{array}$$

We now give several fundamental examples of left/right D_n ($= D_{\mathbb{A}^n}$)-modules. Warning: Since D_n is non-commutative, we must be careful when distinguishing between left and right D modules!

Example 1.11. I. D_n is automatically both a left and right D_n -module. (via the natural action.)

II. $\mathcal{O}_X$ is naturally a left D_X -module via the canonical action of D_X .

III. Given operators $P_1, \dots, P_N \in D_n$, the left (resp. right) ideal $\sum_i D_n P_i$ (resp. $\sum_i P_i D_n$) is a left (resp. right) D_n -modules.

Similarly, the quotient $D_n / \sum_i D_n P_i$ (resp. $D_n / \sum_i P_i D_n$) is a left (resp. right) D_n -module.

IV. Let F be a space of $\mathbb{C}$ -valued functions on $\mathbb{C}^n$ that is an algebra over the polynomial ring and closed under differentiation. It has a left D_n -module structure.

Definition 1.12. Fix a space of functions F as in Ex.8 IV, and let M be a left D_X -module. We define the space of solutions of M by

$$\text{Sol}(M) := \text{Hom}_{D_X}(M, F)$$

At first, this definition appears strange, but here is an example providing motivation.

Example 1.13 (Motivation). Let $P_1, \dots, P_s \in D_{\mathbb{A}^n}$, and consider the left $D_{\mathbb{A}^n}$ -module given by

$$M = D_n / D_n P_1 + \dots + D_n P_s$$

In this case, after a short computation, we find that.

$$\text{Hom}_{D_{\mathbb{A}^n}}(M, \mathcal{O}_{\mathbb{A}^n}) = \{f \in \mathcal{O}_{\mathbb{A}^n} | P_i f = 0 \ (\forall i = 1, \dots, s)\}$$

There is a natural way to think of a connection on a vector bundle as a D -module. It is a vast generalization. We will try to explain the correspondence between D -modules and connections now.

First, we recall what a connection on a locally free sheaf is. We give two equivalent (dual) definitions:

Definition 1.14. Let M be a locally free $\mathcal{O}_X$ -module. A $\mathbb{C}$ -linear connection is a $\mathbb{C}$ -linear morphism

$$\nabla : M \longrightarrow M \otimes_{\mathcal{O}_X} \Omega_X^1$$

satisfying the Leibniz identity $\nabla(fs) = df \otimes s + f\nabla s$.

This extends to $\nabla : \Omega_X^p \otimes M \rightarrow \Omega_X^{p+1} \otimes M$ via $\nabla(\omega \otimes s) = d\omega \otimes s + (-1)^p \omega \otimes \nabla s$.

A connection is called integrable (or flat) if its curvature, $\Theta := \nabla^2 = \nabla \circ \nabla$, vanishes.

Definition 1.15. Let M be a locally free $\mathcal{O}_X$ -module. A connection ∇ on M is a $\mathbb{C}$ -linear morphism

$$\nabla : \Theta_X \longrightarrow \text{End}_{\mathbb{C}}(M) \quad (\theta \mapsto \nabla_\theta)$$

satisfying:

- i. $\nabla_{f\theta}(s) = f\nabla_\theta(s) \quad (f \in \mathcal{O}_X, \theta \in \Theta_X, s \in M)$
- ii. $\nabla_\theta(fs) = \theta(f)s + f\nabla_\theta(s)$

Furthermore, if $\nabla_{[\theta_1, \theta_2]}(s) = [\nabla_{\theta_1}, \nabla_{\theta_2}](s)$, $\forall \theta_1, \theta_2 \in \Theta_X$, then the connection is called integrable.

Proposition 1.16. *Let M be a locally free $\mathcal{O}_X$ -module. A left D_X -module structure on M which extends the $\mathcal{O}_X$ -module structure is equivalent to an integrable connection (c.f. def. 1.15).*

$$\begin{aligned} \nabla : \Theta_X &\longrightarrow \text{End}_{\mathbb{C}}(M) \\ \theta &\longmapsto \nabla_\theta \end{aligned}$$

Proof. Clear by the definitions. □

2 Direct and Inverse Images

Let $f : X \rightarrow Y$ be a morphism of smooth complex varieties. We want to define the inverse image of a D_Y -module M .

2.1 Inverse Images

Definition 2.1. Let M be a left D_Y -module. Its inverse image under f is defined to be

$$f^*M := \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}M,$$

which coincides with the usual inverse image of $\mathcal{O}$ -modules.

We must describe the left D_X -structure on f^*M . The $\mathcal{O}_X$ -structure is given by the usual pullback of $\mathcal{O}$ -modules, so we must understand how Θ_X acts on the inverse image f^*M . To this end, there is a natural map

$$\begin{aligned} \Theta_X &\longrightarrow f^*\Theta_Y = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\Theta_Y \\ \theta &\longmapsto \tilde{\theta} = \sum_j \varphi_j \otimes \theta_j \end{aligned}$$

Geometrically, this map corresponds to the pushforward of tangent vectors. More fundamentally, this is nothing other than the chain rule.

Let $t \otimes m \in f^*M$, we can define the Θ_X action on f^*M as follows:

$$\theta(t \otimes m) := \theta(t) \otimes m + t \otimes \tilde{\theta}(m)$$

We can make this explicit by looking in an open affine neighborhood. Let's assume for simplicity that M is quasi-coherent $\mathcal{O}_Y$ -module, and let $\{y_i, \partial_i\}_{i=1, \dots, n}$ be a local coordinate system on Y . Then, the Θ_X -structure is given by

$$\theta(t \otimes m) := \theta(t) \otimes m + t \sum_{i=1}^n \theta(y_i \circ f) \otimes \partial_i m$$

It is easy to check that this definition is well-defined. Indeed, let $\{y'_i, \partial'_i\}$ be another coordinate system. Then

$$\begin{aligned} t \sum_j \theta(y'_j \circ f) \otimes \partial_j m &= t \sum_{i,j,k} \left(\frac{\partial y'_j}{\partial y_i} \circ f \right) \theta(y_i \circ f) \otimes \left(\frac{\partial y_k}{\partial y_j} \partial_k m \right) \\ &= t \sum_{i,k} \theta(y_i \circ f) \otimes \partial_k m \left(\sum_j \frac{\partial y'_j}{\partial y_i} \frac{\partial_k}{\partial y'_j} \circ f \right) \\ &= t \sum_{i,k} \theta(y_i \circ f) \otimes (\partial_k m) \partial_{ik} \\ &= t \sum_i \theta(y_i \circ f) \otimes \partial_i m \end{aligned}$$

Proposition 2.2. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ morphisms of smooth algebraic varieties, then $(g \circ f)^* = f^* \circ g^*$.*

2.2 First Transfer Bimodule

Let us now consider D_Y as a left D_Y -module, so that $f^*D_Y = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_X} f^{-1}D_Y$ is a left D_X -module. At the same time, it has a natural right $f^{-1}D_Y$ -module structure induced by right multiplication.

Definition 2.3 (Transfer Bimodule). The $(D_X, f^{-1}D_Y)$ -bimodule $f^*D_Y = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_X} f^{-1}D_Y$ is called the transfer bimodule, and is denoted by $D_{X \rightarrow Y}$.

Putting all the definitions together, we may rewrite inverse images as

$$f^*M \cong D_{X \rightarrow Y} \otimes_{f^{-1}D_Y} f^{-1}M$$

as left D_X -modules. So we have the right exact functor

$$D_{X \rightarrow Y} \otimes_{f^{-1}D_Y} f^{-1}(-) : \text{Mod}(D_Y) \rightarrow \text{Mod}(D_X)$$

The main point is that to understand inverse images, it suffices to understand the structure of the transfer bimodule $D_{X \rightarrow Y}$.

Example 2.4. Let $i : X = \mathbb{A}^r \hookrightarrow Y = \mathbb{A}^n$ be the closed embedding. Let $\{y_i, \partial_{y_i}\}_{1 \leq i \leq n}$ be a local coordinate system on Y such that X is given by the system of equations $y_{r+1} = \dots = y_n = 0$ in Y .

So we get a local coordinate system $\{x_i, \partial_{x_i}\}_{1 \leq i \leq r}$ with $x_k = y_k \circ i$ for all $1 \leq k \leq r$. Moreover, the canonical map $\Theta_X \rightarrow f^* \Theta_Y$ is given by $\partial_{x_k} \mapsto \partial_{y_k}$ for $k = 1, \dots, r$. Then,

$$\begin{aligned} D_{X \rightarrow Y} &= \mathbb{C}[\underline{x}] \otimes_{\mathbb{C}[\underline{y}]} \mathbb{C}[\underline{y}, \underline{\partial}_y] \\ &\cong \mathbb{C}[\underline{x}] \otimes_{\mathbb{C}[\underline{y}]} \mathbb{C}[\underline{y}, \partial_{y_1}, \dots, \partial_{y_r}] \otimes_{\mathbb{C}} \mathbb{C}[\partial_{y_{r+1}}, \dots, \partial_{y_n}] \\ &\cong D_X \otimes_{\mathbb{C}} \mathbb{C}[\partial_{y_{r+1}}, \dots, \partial_{y_n}] \end{aligned}$$

The left D_X -action is given by left multiplication.

Remark 2.5. The previous example is locally true for any closed immersion $X \hookrightarrow Y$ of smooth algebraic varieties.

Example 2.6. Let $X = \mathbb{C}^n \xrightarrow{p} Y = \mathbb{C}^{n-1}$ the projection morphism with $p(x_1, \dots, x_n) = (x_1, \dots, x_{n-1})$.

Then, $p^*M = \mathbb{C}[x_n] \otimes_{\mathbb{C}} M$. Given an element $f(x_n) \otimes m \in p^*M$, x_n, ∂_n act on the first factor, and the remaining generators of D_X act on the second factor.

2.3 Direct Images of Right D-modules

Perhaps surprisingly, direct images are more easily defined for the right D -modules, so we shall start with that case. Let $f : X \rightarrow Y$ be a map of smooth algebraic varieties, and let M be a right D_X -module. Our first guess would be to consider the usual pushforward f_*M and try to endow the structure of a right D_Y -module. Unfortunately, that doesn't work. (cf. Ex.)

Example 2.7.

After further analysis, we consider $M \otimes_{D_X} D_{X \rightarrow Y}$, which has a natural right $f^{-1}D_Y$ -module structure. Applying the sheaf theoretic direct image functor f_* , we end up with a right D_Y -module $f_*(M \otimes_{D_X} D_{X \rightarrow Y})$. So,

$$f_*((-) \otimes_{D_X} D_{X \rightarrow Y} : \text{Mod}(D_X^{op}) \rightarrow \text{Mod}(D_Y^{op})$$

Is the direct image functor for the right D modules.

Remark 2.8. The main issue with the above definition is that it mixes a left-exact functor (namely f_*) with a right-exact functor ($\otimes$). This composition introduces significant pathologies in the homological properties of the direct image, which results in the breakdown of many homological arguments. Considering everything in the realm of derived categories will alleviate the previous problem.

We want to explain how to define the direct image of left D -modules. The key idea is that an involution exists between left D -modules and right D -modules. This will allow us to convert a left D_X -module to a right D_X -module and vice-versa. To summarize:

Main Idea:

1. Convert the left D_X -module to a right D_X -module.
2. Take the direct image under f as explained above.
3. Convert it back to a left D_Y -module.

2.4 Correspondence between left and right D-modules.

First, local description. Let $\{x_i, \partial_i\}_{1 \leq i \leq n}$ be a local coordinate system on an open affine $U \subseteq X$. There is an involution $P \mapsto P^*$ of A_n given by $x_i^* = x_i$ and $\partial_i^* = -\partial_i$.

Given a right A_n -module M , we may define a left A_n -module structure on M by

$$P \cdot m := mP^*, \quad (\forall m \in M)$$

We denote this left A_n -module by $Flip^{R \leftarrow R}(M)$. This gives an equivalence between $Mod(A_n^{op})$ and $Mod(A_n)$. The inverse operation will be denoted by $Flip^{L \rightarrow R}(-)$.

The problem with this local description is that it is not at all clear whether this operation glues over affine open subsets to give a global operator $Flip^{R \rightarrow L}(-) : Mod(D_X^{op}) \rightarrow Mod(D_X)$. We need to consider the canonical sheaf to globalize this correspondence to an arbitrary smooth variety X .

$$\omega_X = \bigwedge^{dim X} \Omega_X^1$$

This has a natural right D_X -module structure given by the Lie derivative as follows.

$$\omega\theta := -(\text{Lie}\theta)\omega \quad (\omega \in \Omega_X^n, \theta \in \Theta_X)$$

Where the Lie-derivative is defined by

$$(\text{Lie}(\theta) \cdot \omega)(\theta_1, \dots, \theta_n) = \theta(\omega(\theta_1, \dots, \theta_n)) - \sum_{i=1}^n \omega(\theta_1, \dots, [\theta, \theta_i], \dots, \theta_n)$$

where $\omega \in \Omega_X^n$ and $\theta, \theta_i \in \Theta_X$ for all $i = 1, \dots, n$.

Lemma 2.9. *In a local coordinate system $\{x_i, \partial_i\}$ we can express this right- D_X -module structure as:*

$$(f dx_1 \wedge \dots \wedge dx_n) P(x, \partial) = (P(x, \partial)^* f) dx_1 \wedge \dots \wedge dx_n$$

where $x_i^* = x_i$ and $\partial_i^* = -\partial_i$ as before.

Proof. This lemma is proved by directly computing the Lie derivative. $\square$

The right D_X -module structure on ω_X gives a homomorphism $D_X^{op} \rightarrow \text{End}_{\mathbb{C}}(\omega_X)$ of $\mathbb{C}$ -algebras.

Note: $\text{End}_{\mathbb{C}}(\omega_X) \cong \omega_X \otimes_{\mathcal{O}_X} \text{End}_{\mathbb{C}}(\mathcal{O}_X) \otimes_{\mathcal{O}_X} \omega_X^{\vee}$ of $\mathbb{C}$ -algebras.

Lemma 2.10. *We have a canonical isomorphism*

$$D_X^{op} \cong \omega_X \otimes_{\mathcal{O}_X} D_X \otimes_{\mathcal{O}_X} \omega_X^{\vee}$$

of $\mathbb{C}$ -algebras.

In terms of local coordinates $\{x_i, \partial_i\}$, this isomorphism is explicitly given by

$$P^{\circ} \in D_X^{op} \mapsto d\underline{x} \otimes P^* \otimes d\underline{x}^{\otimes -1} \in D_X$$

2.5 Direct Images of Left D -modules.

Recall that the direct image functor of right D -modules given by

$$f_*((-) \otimes_{D_X} D_{X \rightarrow Y}) : \text{Mod}(D_X^{op}) \rightarrow \text{Mod}(D_Y^{op})$$

where $D_{X \rightarrow Y}$ is the $(D_X, f^{-1}D_Y)$ -bimodule $\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y$. A candidate for the direct image of left D -modules is

$$\begin{array}{ccc} \text{Mod}(D_X) & \dashrightarrow & \text{Mod}(D_Y) \\ \omega_X \otimes_{\mathcal{O}_X} (-) \Big\downarrow \cong & & \omega_Y \otimes_{\mathcal{O}_Y} (-) \Big\downarrow \cong \\ \text{Mod}(D_X^{op}) & \xrightarrow{f_*((-) \otimes_{D_X} D_{X \rightarrow Y})} & \text{Mod}(D_Y^{op}) \end{array}$$

Therefore, it is natural to consider

$$M \mapsto \omega_Y^\vee \otimes_{\mathcal{O}_Y} f_*((\omega_X \otimes_{\mathcal{O}_X} M) \otimes_{D_X} D_{X \rightarrow Y})$$

Where $\omega_Y^\vee$ denotes the dual sheaf.

Now, we have an isomorphism.

$$(\omega_X \otimes_{\mathcal{O}_X} M) \otimes_{D_X} D_{X \rightarrow Y} \cong (\omega_X \otimes_{\mathcal{O}_X} D_{X \rightarrow Y}) \otimes_{D_X} M$$

of right $f^{-1}D_Y$ -modules, where $f^{-1}D_Y$ acts on $(\omega_X \otimes_{\mathcal{O}_X} D_{X \rightarrow Y}) \otimes_{D_X} M$ by

$$((\omega \otimes R) \otimes S)P = (\omega \otimes RP) \otimes s \quad (\omega \in \omega_X, R \in D_{X \rightarrow Y}, s \in M, P \in D_Y)$$

Therefore:

$$\begin{aligned} & \omega_Y^\vee \otimes_{\mathcal{O}_X} f_*((\omega_X \otimes_{\mathcal{O}_X} M) \otimes_{D_X} D_{X \rightarrow Y}) \\ & \cong \omega_Y^\vee \otimes_{\mathcal{O}_Y} f_*((\omega_X \otimes_{\mathcal{O}_X} D_{X \rightarrow Y}) \otimes_{D_X} M) \\ & \cong f_*((\omega_X \otimes_{\mathcal{O}_X} D_{X \rightarrow Y} \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y^\vee) \otimes_{D_X} M) \end{aligned}$$

Definition 2.11. (Second Transfer Bimodule) The $(f^{-1}D_Y, D_X)$ -bimodule $\omega_X \otimes_{\mathcal{O}_X} D_{X \rightarrow Y} \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y^\vee$ is called the transfer bimodule, and is denoted by $D_{Y \leftarrow X}$.

Our tentative direct image functor is given by:

$$f_+(-) = f_*(D_{Y \leftarrow X} \otimes_{D_X} (-)) : \text{Mod}(D_X) \longrightarrow \text{Mod}(D_Y)$$

Lemma 2.12. We have that $D_{Y \leftarrow X} \cong f^{-1}(D_Y \otimes_{\mathcal{O}_X} \omega_Y^\vee) \otimes_{f^{-1}\mathcal{O}_Y} \omega_X$.

Proof. Indeed, we have

$$\begin{aligned} D_{Y \leftarrow X} &:= \omega_X \otimes_{\mathcal{O}_X} D_{X \rightarrow Y} \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y^\vee \\ &\cong \omega_X \otimes_{\mathcal{O}_X} (\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y) \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y^\vee \\ &\cong \omega_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y^\vee \\ &\cong \omega_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}(D_Y \otimes_{\mathcal{O}_Y} \omega_Y^\vee) \\ &\cong \omega_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}(\omega_Y \otimes_{\mathcal{O}_Y} D_Y^{op} \otimes_{\mathcal{O}_Y} \omega_Y \otimes_{\mathcal{O}_Y} \omega_Y^\vee) \\ &\cong \omega_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}(\omega_Y \otimes_{\mathcal{O}_Y} D_Y^{op}) \\ &\cong f^{-1}(D_Y \otimes_{\mathcal{O}_Y} \omega_Y^\vee) \otimes_{f^{-1}\mathcal{O}_Y} \omega_X \end{aligned}$$

□

Example 2.13. Same setup as (...). Let $i : X = \mathbb{A}^r \hookrightarrow Y = \mathbb{A}^n$ closed embedding. Let $\{y_i, \partial_{y_i}\}_{1 \leq i \leq n}$ be a local coordinate system on Y , such that $X = \mathbb{A}^r$ is "cut" by the equations $y_{r+1} = \dots = y_n = 0$.

We have a local coordinate system $\{x_i, \partial_{x_i}\}_{1 \leq i \leq r}$ on X such that $x_k = y_k \circ i$, for all $i = 1, \dots, r$. Then $i^{-1} \omega_Y^\vee \otimes_{i^{-1} \mathcal{O}_Y} \omega_X$ can be identified with $\mathcal{O}_X$ by the section $(dy_1 \wedge \dots \wedge dy_n)^{\otimes -1} \otimes (dx_1 \wedge \dots \wedge dx_r)$. So,

$$\begin{aligned} D_{Y \leftarrow X} &\cong \mathbb{C}[y, \underline{\partial}_y] \otimes_{\mathbb{C}[y]} \mathbb{C}[x] \\ &\cong \mathbb{C}[\partial_{y_{r+1}}, \dots, \partial_{y_n}] \otimes_{\mathbb{C}} D_X. \end{aligned}$$

More explicitly, when $i : X = \mathbb{C}^n \hookrightarrow Y = \mathbb{C}^{n+1}$ and M is a D_X -module, we have that $i_* M = M^{\mathbb{N}}$, a countable direct sum. The action of $x_1, \partial_1, \dots, x_n, \partial_n$ is dictated by the D_X -structure, x_{n+1} acts by 0, and ∂_{n+1} acts as the shift operator

$$(m_1, m_2, \dots) \mapsto (0, m_1, m_2, \dots)$$

In this case, we can write $i_+ M = \bigoplus_j \partial_{n+1}^j M$.

3 Introduction to Derived Categories and Derived Functors

Nope—check Appendix B in HTT.

4 Direct and Inverse Images—Part 2

We start by introducing some new notation.

Notation:

1. $Mod_{qc}(D_X)$: category of quasi-coherent D_X -modules.
2. $Mod_c(D_X)$: category of coherent D_X -modules
3. $D_{qc}^\#(D_X)$ (resp. $D_c^\#(D_X)$): full subcategory of $D^\#(D_X)$ with cohomology sheaves in $Mod_{qc}(D_X)$ (resp. in $Mod_c(D_X)$), where $\# = -, +, b$.

Proposition 4.1. *Let X be a quasi-projective variety, and $M \in Mod_{qc}(D_X)$.*

1. *M has an injective resolution in $Mod_{qc}(D_X)$;*
2. *M has a locally free resolution in $Mod_{qc}(D_X)$;*
3. *M has a finite resolution by locally projective object in $Mod_{qc}(D_X)$ of length $\leq 2 \dim X$.*

4.1 Derived Inverse Images

Definition 4.2. Let $f : X \rightarrow Y$ be a morphism of smooth quasi-projective varieties.

$$\mathbb{L}f^* : D^b(D_Y) \longrightarrow D^b(D_X)$$

$$M^\bullet \longmapsto D_{X \rightarrow Y} \otimes_{f^{-1}D_Y}^{\mathbb{L}} f^{-1}M^\bullet$$

Proposition 4.3. *The functor $\mathbb{L}f^*$ descends to a functor $D_{qc}^b(D_Y) \rightarrow D_{qc}^b(D_X)$.*

Proof. $M^\bullet \in D_{qc}^b(D_Y)$ is quasi-isomorphic to a bounded complex of locally projective objects $P^\bullet$. Therefore, we can compute

$$\begin{aligned} \mathbb{L}f^*M^\bullet &= D_{X \rightarrow Y} \otimes_{f^{-1}D_Y} f^{-1}P^\bullet \\ &\cong (\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y) \otimes_{f^{-1}D_Y} f^{-1}P^\bullet \\ &\cong \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}P^\bullet \cong f^*P^\bullet \end{aligned}$$

the last complex has quasi-coherent cohomology. $\square$

Warning: $\mathbb{L}f^*$ does not send $D_c^b(D_Y)$ to $D_c^b(D_X)$ because $D_{X \rightarrow Y}$ is not coherent.

Definition 4.4. (Shifted Inverse Image) We define the shifted inverse image functor, which will be useful to make many duality arguments work naturally, as well as Kashiwara's equivalence.

$$f^\dagger : D^b(D_Y) \longrightarrow D^b(D_X)$$

$$M^\bullet \longmapsto \mathbb{L}f^*M^\bullet[\dim X - \dim Y]$$

Proposition 4.5. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of smooth quasi-projective varieties X, Y, Z . Then, $\mathbb{L}(g \circ f)^*M^\bullet = \mathbb{L}f^*(\mathbb{L}g^*M^\bullet)$, for any $M^\bullet \in D^b(D_Z)$.*

Proof. It suffices to show that $D_{X \rightarrow Z} \cong D_{X \rightarrow Y} \otimes_{f^{-1}D_Y}^{\mathbb{L}} f^{-1}D_{Y \rightarrow Z}$. Indeed,

$$\begin{aligned} &D_{X \rightarrow Y} \otimes_{f^{-1}D_Y}^{\mathbb{L}} f^{-1}D_{Y \rightarrow Z} \\ &= (\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y) \otimes_{f^{-1}D_Y}^{\mathbb{L}} f^{-1}(\mathcal{O}_Y \otimes_{g^{-1}\mathcal{O}_Z} g^{-1}D_Z) \\ &\cong (\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y) \otimes_{f^{-1}D_Y}^{\mathbb{L}} f^{-1}\mathcal{O}_Y \otimes_{(g \circ f)^{-1}\mathcal{O}_Z} (g \circ f)^{-1}D_Z \\ &\cong \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y}^{\mathbb{L}} f^{-1}D_Y \otimes^{\mathbb{L}} (g \circ f)^{-1}D_Z \\ &\cong \mathcal{O}_X \otimes_{(g \circ f)^{-1}\mathcal{O}_Z}^{\mathbb{L}} (g \circ f)^{-1}D_Z \\ &= D_{X \rightarrow Z}. \end{aligned}$$

□

Example 4.6. Let $j : U \hookrightarrow X$ be an open embedding into a smooth algebraic variety X . Then, j^{-1} is just the restriction to U , and in particular $D_{U \rightarrow X} = \mathcal{O}_U \otimes_{f^{-1}\mathcal{O}_X} f^{-1}D_X = D_U$, and $j^\dagger = \mathbb{L}j^* = j^{-1}$.

4.2 Box Products and Tensor Products

The tensor bifunctor

$$(-) \otimes_{\mathcal{O}_X} (-) : \text{Mod}(D_X) \times \text{Mod}(D_X) \rightarrow \text{Mod}(D_X) \quad ((M, N) \mapsto M \otimes_{\mathcal{O}_X} N)$$

Is right exact and, using flat resolutions, we may define its left-derived functor

$$(-) \otimes_{\mathcal{O}_X}^{\mathbb{L}} (-) : D^b(D_X) \times D^b(D_X) \rightarrow D^b(D_X) \quad ((M^\bullet, N^\bullet) \mapsto M^\bullet \otimes_{\mathcal{O}_X}^{\mathbb{L}} N^\bullet)$$

Let X and Y be smooth algebraic varieties and let $pr_1 : X \times Y \rightarrow X$, $pr_2 : X \times Y \rightarrow Y$ be the first and second projection morphisms. It turns out that $\Theta_{X \times Y} \cong pr_1^* \Theta_X \oplus pr_2^* \Theta_Y$. This raises the question: do we have a similar isomorphism relating $D_{X \times Y}$ to D_X and D_Y ? The answer is affirmative based on the following lemma:

Lemma 4.7. $D_{X \times Y} \cong \mathcal{O}_{X \times Y} \otimes_{pr_1^{-1}\mathcal{O}_X \otimes_{\mathbb{C}} pr_2^{-1}\mathcal{O}_Y} (pr_1^{-1}D_X \otimes_{\mathbb{C}} pr_2^{-1}D_Y)$

This isomorphism motivates the following definition.

Definition 4.8. Let $M \in \text{Mod}(D_X)$ and $N \in \text{Mod}(D_Y)$, we define the box product of M and N to be

$$M \boxtimes N := \mathcal{O}_{X \times Y} \otimes_{pr_1^{-1}\mathcal{O}_X \otimes_{\mathbb{C}} pr_2^{-1}\mathcal{O}_Y} (pr_1^{-1}M \otimes_{\mathbb{C}} pr_2^{-1}N) \in \text{Mod}(\mathcal{O}_{X \times Y})$$

The box product $(-) \boxtimes (-)$ is exact on both factors, so it defines a derived functor.

$$(-) \boxtimes (-) : D^b(D_X) \times D^b(D_Y) \rightarrow D_{X \times Y}^b$$

Which preserves both quasi-coherence and coherence.

Proposition 4.9. *i) Let $f_1 : X_1 \rightarrow Y_i$ ($i = 1, 2$) be morphisms of smooth algebraic varieties, and $M_1^\bullet \in D^b(D_{Y_1})$, $M_2^\bullet \in D^b(D_{Y_2})$. Then,*

$$\mathbb{L}(f_1 \times f_2)^*(M_1^\bullet \boxtimes M_2^\bullet) \cong (\mathbb{L}f_1^* M_1^\bullet) \boxtimes (\mathbb{L}f_2^* M_2^\bullet)$$

ii) Let $f : X \rightarrow Y$ be a morphism of smooth algebraic varieties and $M^\bullet, N^\bullet \in D^b(D_Y)$. Then,

$$\mathbb{L}f^*(M_1 \otimes_{\mathcal{O}_Y}^{\mathbb{L}} M_2) \cong \mathbb{L}f^*M_1 \otimes_{\mathcal{O}_X}^{\mathbb{L}} \mathbb{L}f^*M_2$$

We continue our discussion on inverse images. We first treat the case where we have a smooth projection.

Smooth Projections. Let $pr_1 : X \times Y \rightarrow X$ and $pr_2 : X \times Y \rightarrow Y$ be the usual projections. Then, it is easy to check that

$$pr_1^*M \cong M \boxtimes \mathcal{O}_Y, \quad pr_2^*N \cong \mathcal{O}_X \boxtimes N$$

Since $-\boxtimes-$ is exact, we get that

$$pr_1^\dagger M \cong \mathbb{L}pr_1^*M^\bullet[\dim Y], \text{ and as a result } \\ H^i(pr_1^\dagger M^\bullet) = pr_1^*(H^{i+\dim Y}(M^\bullet))$$

Proposition 4.10. *If $f : X \rightarrow Y$ is a smooth morphism of smooth algebraic varieties, and $M \in Mod(D_Y)$. Then, $H^i(\mathbb{L}f^*M) = 0$ for all $i \neq 0$, and if $M \in Mod_c(D_Y)$, then $\mathbb{L}f^*M \in Mod_c(D_X)$.*

Proof. The first assertion follow from the fact that $\mathcal{O}_X$ is a flat $f^{-1}\mathcal{O}_Y$ -module, and using that $\mathbb{L}f^*M = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y}^{\mathbb{L}} f^{-1}M$.

For the second assertion, it suffices to show that the natural map $D_X \rightarrow D_{X \rightarrow Y}$ is surjective. Indeed, working on a local coordinate system, we calculate that.

$$D_{X \rightarrow Y} = \bigoplus_{r_1, \dots, r_m} \mathcal{O}_X \partial_{y_1}^{r_1} \dots \partial_{y_m}^{r_m}$$

and $D_X \rightarrow D_{X \rightarrow Y}$ is given by $\partial_{x_1}^{r_1} \dots \partial_{x_n}^{r_n} \mapsto \delta_{r_{m+1}+\dots+r_n, 0} \partial_{y_1}^{r_1} \dots \partial_{y_m}^{r_m}$. $\square$

Next, we discuss the case of closed embeddings.

Closed Embeddings. Let $i : X \hookrightarrow Y$ be a closed embedding of smooth quasi-projective varieties and denote by $d = \dim Y - \dim X$. Since X, Y are smooth, the image of X under i in Y is a local complete intersection (lci for short). We want to describe the complex $\mathbb{L}i^*M$, where $M \in Mod_{qc}(D_X)$. Recall that

$$\mathbb{L}i^*M := D_{X \rightarrow Y} \otimes_{i^{-1}\mathcal{D}_Y}^{\mathbb{L}} i^{-1}M$$

It suffices to find a locally free resolution for $D_{X \rightarrow Y}$. To do this, we use the Koszul complex.

Since X is a lci in Y , locally, the ideal sheaf I_X is defined by a regular sequence $y_1, \dots, y_d = 0$. Locally, we have a Koszul resolution

$$0 \rightarrow K_d \rightarrow \dots \rightarrow K_1 \rightarrow K_0 \rightarrow \mathcal{O}_X \rightarrow 0$$

of the $f^{-1}\mathcal{O}_Y$ -module $\mathcal{O}_X$, where

$$K_p := \bigwedge^p \bigoplus_{j=1}^d i^{-1}\mathcal{O}_Y \cdot dy_j$$

and

$$\begin{aligned} d : K_p &\longrightarrow K_{p-1} \\ dy_{i_1} \wedge \dots \wedge dy_{i_p} &\mapsto \sum_{j=1}^p (-1)^{j+1} y_{i_j} dy_{i_1} \wedge \dots \wedge \widehat{dy_{i_j}} \wedge \dots \wedge dy_{i_p} \end{aligned}$$

Lemma 4.11. *There exists a global locally-free resolution of the right $i^{-1}D_Y$ -module $D_{X \rightarrow Y}$*

$$0 \rightarrow K_d \otimes_{i^{-1}\mathcal{O}_Y} i^{-1}D_Y \rightarrow \dots \rightarrow K_0 \otimes_{i^{-1}\mathcal{O}_Y} i^{-1}D_Y \rightarrow D_{X \rightarrow Y} \rightarrow 0$$

and so $\mathbb{L}i^*M$ is represented by the complex

$$\mathbb{L}i^*M = [K_d \otimes_{i^{-1}\mathcal{O}_Y} i^{-1}M \rightarrow \dots \rightarrow K_0 \otimes_{i^{-1}\mathcal{O}_Y} i^{-1}M]$$

Proof. Tensor the Koszul complex by $i^{-1}D_Y$, which is flat over $i^{-1}\mathcal{O}_Y$. $\square$

Proposition 4.12. *Let $i : X \hookrightarrow Y$ as above and $d = \dim X - \dim Y$.*

i) If $M \in \text{Mod}(D_Y)$, then $H^p(i^\dagger M) = 0$, unless $0 \leq p \leq d$. Moreover,

$$\begin{aligned} H^0(i^\dagger M) &\cong \omega_{X/Y}^{-1} \otimes_{\mathcal{O}_X} \{m \in M \mid I_X \cdot m = 0\} \\ H^d(i^\dagger M) &\cong M/I_X M \end{aligned}$$

ii) For $M^\bullet \in D^+(D_Y)$, we have a canonical isomorphism

$$\begin{aligned} i^\dagger M^\bullet &\cong \mathbb{R}\mathcal{H}om_{i^{-1}D_Y}(D_{Y \leftarrow X}, i^{-1}M^\bullet)[d] \\ &\cong \mathbb{R}\mathcal{H}om_{i^{-1}D_Y}(D_{Y \leftarrow X}, i^{-1}\Gamma_X(M^\bullet))[d] \end{aligned}$$

where $\Gamma_X(M^\bullet)$ is the subsheaf of M consisting of sections with support in X .

Proof. i) This follows from analyzing the Koszul complex. We skip this part.
ii) (Check HTT pg. 36). $\square$

Definition 4.13. For a closed embedding $i : X \hookrightarrow Y$ of smooth algebraic varieties, define a left exact functor.

$$i^{\natural} : \text{Mod}(D_Y) \rightarrow \text{Mod}(D_X)$$

by $i^{\natural}M = \text{Hom}_{i^{-1}D_Y}(D_{Y \leftarrow X}, i^{-1}M)$.

4.3 Derived Direct Images

Definition 4.14. Let $f : X \rightarrow Y$ be a map of smooth algebraic varieties, and $M^{\bullet} \in D^b(D_X)$. We define the derived direct image by

$$\int_f : D^b(D_X) \rightarrow D^b(D_Y); \quad \int_f M^{\bullet} := \mathbb{R}f_*(D_{Y \leftarrow X} \otimes_{D_X}^{\mathbb{L}} M^{\bullet})$$

We will also use the following notation.

$$\int_f^k M^{\bullet} := H^k \left(\int_f M^{\bullet} \right)$$

We start with two lemmas.

Lemma 4.15. *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Then*

$$D_{Z \leftarrow X} \cong f^{-1}D_{Z \leftarrow Y} \otimes_{f^{-1}D_Y} D_{Y \leftarrow X}$$

Lemma 4.16 (Projection formula for D -modules.). *If $F^{\bullet} \in D_{qc}^-(D_Y^{op})$ and $G^{\bullet} \in D_{qc}^-(f^{-1}D_Y)$. Then, the natural map*

$$F^{\bullet} \otimes_{D_Y}^{\mathbb{L}} \mathbb{R}f_*(G^{\bullet}) \rightarrow \mathbb{R}f_*(f^{-1}F^{\bullet} \otimes_{f^{-1}D_Y}^{\mathbb{L}} G^{\bullet})$$

is an isomorphism

Proof. The assertion is local, so we may assume Y is affine, and we may replace $F^{\bullet}$ with a complex of free D_Y^{op} modules in $D_{qc}^-(D_Y^{op})$ —that is, $F^{\bullet} = D_Y^{\oplus I}$, for some (possibly) infinite index I . Then,

$$F^{\bullet} \otimes_{D_Y}^{\mathbb{L}} \mathbb{R}f_*(G^{\bullet}) \cong \mathbb{R}f_*(G^{\bullet})^{\oplus I}$$

$$\mathbb{R}f_*(f^{-1}F^{\bullet} \otimes_{f^{-1}D_Y}^{\mathbb{L}} G^{\bullet}) \cong \mathbb{R}f_*((G^{\bullet})^{\oplus I})$$

and $\mathbb{R}f_*(-)$ commutes with arbitrary direct sums. $\square$

This leads us to the following proposition

Proposition 4.17 (Composition Rule.). *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of smooth algebraic varieties. We have an isomorphism of functors.*

$$\int_g \int_f \cong \int_{g \circ f} : D^b(D_X) \longrightarrow D^b(D_Z)$$

Proof.

$$\begin{aligned} \int_g \int_f M^\bullet &:= \mathbb{R}g_*(D_{Z \leftarrow Y} \otimes_{D_Y}^{\mathbb{L}} \mathbb{R}f_*(D_{Y \leftarrow X} \otimes_{D_Y}^{\mathbb{L}} M^\bullet)) \\ &\cong \mathbb{R}g_* \mathbb{R}f_*(f^{-1} D_{Z \leftarrow Y} \otimes_{f^{-1} D_Y}^{\mathbb{L}} (D_{Y \leftarrow X} \otimes_{D_Y}^{\mathbb{L}} M^\bullet)) \quad (\text{Lemma 4.16}) \\ &\cong \mathbb{R}(g \circ f)_*(D_{Z \leftarrow X} \otimes_{D_X}^{\mathbb{L}} M^\bullet) \quad (\text{Lemma 4.15}) \\ &= \int_{g \circ f} M^\bullet \end{aligned}$$

which is what we wanted to show. $\square$

Example 4.18. Let $j : U \hookrightarrow X$ be an open embedding. Then $D_{U \leftarrow X} = j^{-1} D_X = D_U$ and $\int_j = \mathbb{R}j_*$.

Example 4.19. Let $i : X \hookrightarrow Y$ be a closed embedding of smooth algebraic varieties. Take $\{y_i, \partial_{y_i}\}_{1 \leq i \leq n}$ a local coordinate system such that $y_{r+1} = \dots = y_n = 0$ defines X locally. Then, as we have previously seen (ex. 2.13)

$$\int_i^k M = 0 \quad (k \leq 0), \quad \int_i^0 M \cong \mathbb{C}[\partial_{y_{r+1}}, \dots, \partial_{y_n}] \otimes_{\mathbb{C}} i_* M$$

Moreover, $\int_i^0$ is an exact functor and preserves quasi-coherence. (cf. Kashiwara's equivalence)

Next, we want to describe what $\int_f$ looks like for a smooth projection $f : X = Y \times Z \rightarrow Y$. To that end, we will use the following locally free resolutions.

Lemma 4.20. *We have locally free resolutions for the left D_X -module $\mathcal{I}_X$ and the right D_X -module ω_X :*

$$\begin{aligned} 0 \rightarrow D_X \otimes_{\mathcal{O}_X} \bigwedge^n \Theta_X \rightarrow \dots \rightarrow D_X \otimes_{\mathcal{O}_X} \bigwedge^0 \Theta_X \rightarrow \mathcal{O}_X \rightarrow 0 \\ 0 \rightarrow \Omega_X^0 \otimes_{\mathcal{O}_X} D_X \rightarrow \dots \rightarrow \Omega_X^n \otimes_{\mathcal{O}_X} D_X \rightarrow \omega_X \rightarrow 0 \end{aligned}$$

where $n = \dim X$ with

$$\begin{aligned} d : D_X \otimes_{\mathcal{O}_X} \bigwedge^k \Theta_X &\longrightarrow D_X \otimes_{\mathcal{O}_X} \bigwedge^{k-1} \Theta_X \\ P \otimes \theta_1 \wedge \dots \wedge \theta_k &\longmapsto \sum_i (-1)^{i+1} P \theta_i \otimes \theta_1 \wedge \dots \wedge \widehat{\theta_i} \wedge \dots \wedge \theta_k \\ &\quad + \sum_{i < j} (-1)^{i+j} P \otimes [\theta_i, \theta_j] \wedge \theta_1 \wedge \dots \wedge \widehat{\theta_i} \wedge \dots \wedge \widehat{\theta_j} \wedge \dots \wedge \theta_k \end{aligned}$$

and

$$\begin{aligned} d : \Omega_X^k \otimes_{\mathcal{O}_X} D_X &\longrightarrow \Omega_X^{k+1} \otimes_{\mathcal{O}_X} D_X \\ \omega \otimes P &\longmapsto d\omega \otimes P + \sum_i dz_i \wedge \omega \otimes \partial_i P \end{aligned}$$

Proof. Check HTT pg. 44-45. $\square$

Let Y and Z be smooth algebraic varieties and set $X = Y \times Z$. Let $f : X \rightarrow Y$ and $g : X \rightarrow Z$ be the first and second projections, respectively. We want to describe

$$\int_f M := \mathbb{R}f_*(D_{Y \leftarrow X} \otimes_{D_X}^{\mathbb{L}} M), \quad \text{for } M \in \text{Mod}_{qc}(D_X)$$

To compute $D_{Y \leftarrow X} \otimes_{D_X}^{\mathbb{L}} M$ we use the resolution of the right D_X module $D_{Y \leftarrow X} = D_Y \boxtimes \omega_Z$ induced by the resolution of ω_Z of the previous lemma. Denote by n the relative dimension $\dim X - \dim Y = \dim Z$ and $\Omega_{X/Y}^k = \mathcal{O}_Y \boxtimes \Omega_Z^k$ for $k = 0, \dots, n$ the sheaf of relative differentials. For $M \in \text{Mod}_{qc}(D_X)$ we define its relative de Rham complex

$$(DR_{X/Y}(M))^k := \begin{cases} \Omega_{X/Y}^{n+k} \otimes_{\mathcal{O}_X} M & (-n \leq k \leq 0), \\ 0, & \text{otherwise} \end{cases}$$

with

$$d(\omega \otimes s) = d\omega \otimes s + \sum_{i=1}^n (dz_i \wedge \omega) \otimes \partial_i s.$$

where $\{z_i, \partial_i\}_{1 \leq i \leq n}$ is a local coordinate system of Z . The de Rham complex is naturally a $f^{-1}D_Y$ -module via

$$P(\omega \otimes s) = \omega \otimes ((P \otimes 1)s) \quad (P \in f^{-1}D_Y, \omega \in g^{-1}\Omega_Z^{n+k}, s \in M),$$

And we have that

$$D_{Y \leftarrow X} \otimes_{D_X}^{\mathbb{L}} M \cong DR_{X/Y}(M)$$

in the derived category of $f^{-1}D_Y$ -modules.

Proposition 4.21. *In the same setting as above:*

- i) *For $M \in \text{Mod}(D_X)$ we have $\int_f M \cong \mathbb{R}f_*(DR_{X/Y}(M))$.*
- ii) *Moreover, $\int_f^j M = 0$ unless $-n \leq j \leq n$.*
- iii) *The functor $\int_f$ preserves quasi-coherence.*

Proof. The first assertion follows from the discussion above. The second follows from the fact that f_* has cohomological dimension n together with assertion (i). The final assertion follows from the fact that $DR_{X/Y}(M)^k$ is quasi-coherent $\mathcal{O}_X$ -module. $\square$

Proposition 4.22. *The canonical morphism*

$$\left(\int_{f_1} M_1^\bullet \right) \boxtimes \left(\int_{f_2} M_2^\bullet \right) \rightarrow \int_{f_1 \times f_2} (M_1^\bullet \boxtimes M_2^\bullet)$$

is an isomorphism.

Proposition 4.23 (Base Change Theorem). *For two morphisms $f : Y \rightarrow X$, $g : Z \rightarrow X$ of algebraic varieties consider the fiber product*

$$\begin{array}{ccc} Y \times_X Z & \xrightarrow{\tilde{g}} & Y \\ \tilde{f} \downarrow & & \downarrow f \\ Z & \xrightarrow{g} & X \end{array}$$

Assume that all varieties X , Y , Z , and $Y \times_X Z$ are smooth. Then, there exists an isomorphism

$$g^\dagger \int_f \cong \int_{\tilde{f}} \tilde{g}^\dagger : D_{qc}^b(D_Y) \rightarrow D_{qc}^b(D_Z)$$

Proposition 4.24 (Projection Formula). *Let $f : X \rightarrow Y$ be a morphism of smooth algebraic varieties. Let $M \in D_{qc}^b(D_X)$, $N \in D_{qc}^b(D_Y)$, then*

$$\int_f (M \otimes_{\mathcal{O}_X}^{\mathbb{L}} \mathbb{L}f^* N) \cong \left(\int_f M \right) \otimes_{\mathcal{O}_Y}^{\mathbb{L}} N$$

5 Kashiwara's Equivalence

Theorem 5.1. *Let $i : X \hookrightarrow Y$ be a closed embedding.*

i) *The functor $\int_i^0$ induces an equivalence of categories*

$$\text{Mod}_*(D_X) \longrightarrow \text{Mod}_*^X(D_Y), \text{ where } * = qc \text{ or } c$$

of abelian categories with quasi-inverse $i^\natural = H^0 i^\dagger$.

ii) *For $N \in \text{Mod}_{qc}^X(D_Y)$ we have $H^j i^\dagger N = 0$ for all $j \neq 0$.*

Proof. To show i) it suffices to show that the canonical morphisms

$$M \mapsto i^\natural \int_i^0 M, \quad \int_i^0 i^\natural N \mapsto N$$

are isomorphism. Furthermore, we must argue that $i^\natural$ and $\int_i^0$ preserve quasi-coherence (resp. coherence).

The problem is local, so we may consider everything in local coordinates. Furthermore, we may induct on the codimension of X in Y , reducing the problem to the case where X is hypersurface in Y . Under these assumptions, let $\{y_k, \partial_{y_k}\}_{1 \leq k \leq n}$ be a local coordinate system on Y , such that $\{y_n = 0\} = X$. Set $y = y_n$, $\partial = \partial_{y_n}$, and $\theta = y\partial$. By example 4.18 and prop. 4.12, we get that

$$H^0 i^\dagger N = i^\natural N = \ker\{y : i^{-1}N \rightarrow i^{-1}N\} \quad (4.12)$$

$$H^1 i^\dagger N = \text{coker}\{y : i^{-1}N \rightarrow i^{-1}N\} \quad (4.12)$$

$$H^j i^\dagger N = 0, \text{ when } j \neq 0, 1 \quad (4.12)$$

$$\int_i^0 M = \mathbb{C}[\partial] \otimes_{\mathbb{C}} i_* M \quad (4.19)$$

Consider the eigenspaces $N^j := \{s \in N \mid \theta s = js\}$, where $j \in \mathbb{Z}$. It is easy to check that $yN^j \subseteq N^{j+1}$ and $\partial N^j \subseteq N^{j-1}$. Moreover, θ induces an isomorphism given by multiplication by $j \times - : N^j \xrightarrow{\cong} N^j$, for all $j \neq 0$. As a result, $\partial y = \theta + 1 : N^j \rightarrow N^j$ is an isomorphism when $j \neq -1$. So, when $j < -1$, both maps are isomorphism as seen below

$$\begin{array}{ccccccc} & & & \cong & & & \\ & & & \curvearrowright & & & \\ N^j & \xrightarrow{y} & N^{j+1} & \xrightarrow{\partial} & N^j & \xrightarrow{y} & N^{j+1} \\ & & & \curvearrowleft & & & \\ & & & \cong & & & \end{array}$$

Claim: $N = \sum_{i=1}^{\infty} N^{-i}.$

Assuming the validity of this claim, we immediately get that $H^1 i^\dagger N = 0$, and so ii) is proven. Moreover, by the claim we have that $N = \mathbb{C}[\partial] \otimes_{\mathbb{C}} N^{-1}$ and $i^\sharp N = i^{-1} N^{-1}$. So $\int_i^0 i^\sharp N = \mathbb{C}[\partial] \otimes_{\mathbb{C}} N^{-1} = N$, thus $\text{Mod}_{qc}(D_X) \cong \text{Mod}_{qc}^X(D_Y)$.

Finally, it remains to prove that $\int_i^0$ and $i^\sharp$ preserve coherence. If M is a finitely generated module over D_X , then $\int_i^0 M = \mathbb{C}[\partial] \otimes_{\mathbb{C}} i_* M$ is finitely generated over D_Y .

Assume that $N \in \text{Mod}_{qc}^X(D_Y)$ is f.g. over D_Y . By our claim, $N = \mathbb{C}[\partial] \otimes_{\mathbb{C}} N^{-1}$, so N is generated by finitely many sections $s_1, \dots, s_r$ of N^{-1} . Then $i^\sharp N = i^{-1} N^{-1}$ is generated over D_X by the sections $s_1, \dots, s_r$. The proof is complete. $\square$

Now, we shall prove the claim used to prove Kashiwara's equivalence.

Proof of Claim. Since N is a quasi-coherent $\mathcal{O}_Y$ -module supported on X , any section s is killed by some y^r for $r \gg 0$. Therefore, it suffices to show that

$$\ker\{y^k \times - : N \rightarrow N\} \subseteq \sum_{j=1}^{\infty} N^{-j} \quad (\forall k \geq 1)$$

This formulation allows us to induct on k .

$k = 1$: This case is true since $ys = 0 \implies \theta s = (\partial y - 1)s = -1$.

$k > 1$: and assume that the inclusion holds for $k - 1$. Let $s \in \ker(y^k)$, then

$$y^k s = 0 \implies y^{k-1}(ys) = 0 \implies ys \in \sum_{j=1}^{\infty} N^{-j}$$

Hence $\partial ys \in \sum_{j=2}^{\infty} N^{-j}$, and, as a result,

$$(\theta + 1)s = \theta s + s = y\partial s + s = \partial ys \in \sum_{j=2}^{\infty} N^{-j} \quad (*)$$

On the other hand, $y^{k-1}(\theta s + ks) = \partial(y^k s) = 0$. Therefore, by induction hypothesis

$$\theta s + ks \in \sum_{j=1}^{\infty} N^{-j} \quad (**)$$

Comparing $(*)$ and $(**)$, we conclude that $s \in \sum_{j=1}^{\infty} N^{-j}$. $\square$

6 Coherent D-modules

Good Filtration: Endow D_X with the order filtration $\{F_i D_X\}_{i \in \mathbb{Z}}$. Recall that we may identify $gr^F D_X$ with $\pi_* \mathcal{O}_{T^*X}$. Let M be a D_X -modules, quasi-coherent over $\mathcal{O}_X$. A filtration on M compatible with $\{F_\bullet D_X\}$ is a collection $\{F_i M\}_{i \in \mathbb{Z}}$ satisfying:

1. $F_i M \subseteq F_{i+1} M$ (increasing filtration)
2. $F_i M = 0$ for $i \ll 0$ (bounded below)
3. $M = \bigcup_{i \in \mathbb{Z}} F_i M$ (exhaustive)
4. $(F_j D_X)(F_i M) \subseteq F_{i+j} M$ (compatible with the order filtration)

The graded module $gr^F M$ is a graded- $\pi_* \mathcal{O}_{T^*X}$ -module, quasi-coherent over $\mathcal{O}_X$.

Proposition 6.1. *The following are equivalent:*

1. $gr^F M$ is coherent over $\pi_* \mathcal{O}_{T^*X}$;
2. $F_i M$ is $\mathcal{O}_X$ -coherent and there exists an $i_0 \gg 0$ such that $(F_j D_X)(F_i M) = F_{j+i} M$ ($j \geq 0, i \geq i_0$);
3. There exists a locally surjective D_X -linear morphism $\Phi : D_X^{\oplus m} \rightarrow M$ and integers n_j ($j = 1, \dots, m$), such that

$$\Phi(F_{i-n_1} D_X \oplus \dots \oplus F_{i-n_m} D_X) = F_i M \quad (i \in \mathbb{Z})$$

Definition 6.2. If any of the previous equivalent conditions hold for a filtration $\{F_i M\}$, we call that filtration **good**.

6.1 Characteristic Varieties.

Let X and M be as above, and let $\{F_i M\}$ be a good filtration. Then,

$$\begin{aligned} Ch(M) &:= \text{supp } gr^{\tilde{F}} M \\ &= \text{supp } [\mathcal{O}_{T^*X} \otimes_{\pi^{-1} \pi_* \mathcal{O}_{T^*M}} \pi^{-1} gr^F M] \end{aligned}$$

is called the **characteristic variety** of M . (sometime also denoted by $SS(M)$ for singular support)

Theorem 6.3. *i) Let M be a coherent D_X -module. Then $Ch(M)$ does not depend on the choice of good filtration $F_\bullet M$.*

ii) For a short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of coherent D_X -modules, we have

$$Ch(M) = Ch(M') \cup Ch(M'')$$

Example 6.4. Let (M, ∇) be an integrable connection of rank $r > 0$. Set $F_I M = 0$ ($i < 0$) and $F_i M = M$ for all other values of i . Then $F_\bullet M$ is a good filtration on M , and $gr^F M \cong M \cong \mathcal{O}_X^r$ holds locally.

Moreover, since $\Theta_X \subset \text{ann}_{\pi_* \mathcal{O}_{T^*X}}(gr^F M)$, we conclude that $Ch(M) = T_X^* X \cong X$, where $T_X^* X$ is the zero section of the cotangent bundle.

6.2 Bernstein's Inequality

Let $M \in \text{Mod}_C(D_X)$ and let Λ be an irreducible component of $Ch(M)$, then $\dim \Lambda \geq d_X$, where $d_X := \dim X$.

Proof. We begin by reducing the case where $Ch(M)$ has pure dimension. From the general theory of filtrations, it follows that there exists filtration

$$0 = C^{2d_X+1} \subset C^{2d_X} \subset \dots \subset C^0 = M$$

with the property that $Ch(C^s/C^{s+1})$ has pure codimension s .

Next, we induct on $\dim X$. If $d_X = 0$, then there is nothing to show. Let's assume that $d_X > 0$ and that the statement holds for $k = d_X - 1$. If $\text{supp } M = X$, then $Ch(M) \supset T_X^* X$, so $\dim Ch(M) \geq d_X$. Therefore, we may assume that $\text{supp } M$ is a proper closed subset of X . Moreover, since the question is local, we can further assume that $\text{supp } M$ is contained in some hypersurface $i : H \hookrightarrow X$.

Kashiwara's equivalence implies that there exists a coherent D_H -module N such that $\int_i^0 N = M$. We then have that $Ch(M) = \varpi_i \rho_i^{-1}(Ch(N))$, where ϖ_i and ρ_i

$$T^*X \xleftarrow{\varpi_i} H \times_X T^*X \xrightarrow{\rho_i} T^*H$$

are the natural maps induced by i .

Claim: When i is a closed embedding as above, then $Ch(N) = \varpi_i \rho_i^{-1}(Ch(M))$.

Assuming the validity of the claim, the proof of Bernstein's inequality may easily be completed by noting that $\dim Ch(M) = \dim Ch(N) + 1$ and using induction. $\square$

Proof of claim. The statement of the claim is local. Let $\{x_1, \partial_i\}_{1 \leq i \leq n}$ be a local coordinate system on X , and let $i : H \hookrightarrow X$ be a hypersurface given by $x_1 = 0$. Denote by $x = x_1$ and $\partial = \partial_1$.

Then $M \cong \mathbb{C}[\partial] \otimes_{\mathbb{C}} i_* N$. Let $G_{\bullet} M$ be a good filtration with $G_{-1} = 0$, and define a filtration on N by

$$F_j N = \sum_{k=0}^j \sum_{l \leq j-k} \mathbb{C} \partial^k \otimes_{\mathbb{C}} i_* G_{j-l}(M)$$

It is easy to check that $F_{\bullet} N$ is also a good filtration and

$$gr^F N \cong \bigoplus_{k=0}^j \mathbb{C} \partial^k \otimes_{\mathbb{C}} i_*(G_{j-k} M / G_{j-k-1} M)$$

Therefore

$$\begin{aligned} gr^F N &\cong \mathbb{C}[\xi] \otimes_{\mathbb{C}} gr^G M \\ &\cong \left(\frac{\mathbb{C}[x, \xi]}{x\mathbb{C}[x, \xi]} \right) \otimes_{\mathbb{C}} gr^G M \end{aligned}$$

Hence

$$\begin{aligned} Ch(N) &= \widetilde{\text{supp } gr^F N} = \varpi_i \rho_i^{-1} \widetilde{\text{supp } gr^G M} \\ &= \varpi_i \rho_i^{-1} Ch(M) \end{aligned}$$

□

6.3 Holonomic D-modules

Definition 6.5. A D_X -module M is called **holonomic** if $\dim Ch(M) = \dim X$ or $M = 0$. We denote by $Mod_h(D_X)$ the full subcategory of $Mod_c(D_X)$ consisting of holonomic D_X -modules.

Proposition 6.6. *i) For $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ in $Mod_c(D_X)$ we have M holonomic $\iff M', M''$ are holonomic.*

ii) $Mod_h(D_X)$ is an Artinian category. (This means that any object has finite length.)

Proposition 6.7. *For a non-zero D_X -module M TFAE:*

- I. M is an integrable connection;*
- II. M is coherent over $\mathcal{O}_X$;*
- III. $Ch(M) = T_X^* X$.*

Proof. We have already shown that $I. \iff II.$ and that $I. \implies III.$ (6.4), so we just have to prove that $III. \implies II.$ To that end, the problem is local, so we may pick coordinates $\{x_i, \partial_i\}_{1 \leq i \leq n}$ on X such that $T^*X = X \times \mathbb{C}^n$. Assume that $Ch(M) = T_X^*X$.

This means that for a good filtration $F_\bullet M$, we have

$$\sqrt{ann_{\mathcal{O}_X[\xi_1, \dots, \xi_n]} gr^F M} = \sum_{i=1}^n \mathcal{O}_X[\xi] \xi_i$$

where ξ_i is the principal symbol of ∂_i and we identify $\mathcal{O}_X[\xi]$ with $\pi_* \mathcal{O}_{T^*X}$.

Let $I = \sum_{i=1}^n \mathcal{O}_X[\xi] \xi_i$. Since the ideal I is noetherian, there exists $n_0 \gg 0$ with $I^{n_0} \subset ann_{\mathcal{O}_X[\xi_1, \dots, \xi_n]} gr^F M$. Therefore $\partial^\alpha F_j M = F_{j+n_0-1} M$ where $|\alpha| = n_0$.

At the same time—since $F_\bullet M$ is a good filtration (6.1)—we have $F_i D_X \cdot F_j M = F_{i+j} M$ for $j \gg 0$. It follows that

$$F_{n_0+j} M = (F_{n_0} D_X)(F_j M) \subset F_{j+n_0-1} M$$

and $F_\bullet M$ is exhaustive. Whence $F_\bullet M$ eventually stabilizes and is equal to M . But since $F_\bullet M$ was assumed to be a good filtration, this implies that M is $\mathcal{O}_X$ -coherent (6.1). $\square$

Proposition 6.8. *Let M be a holonomic D_X -module. Then, there exists an open dense subset $U \subseteq X$ such that $M|_U$ is an integrable connection.*

Proof. Set $S = Ch(M) \setminus T_X^*X$. By 6.7, it suffices to find a Zariski open U such that $S|_U = \emptyset$.

If $S = \emptyset$, the proof is complete. Let's assume that $S \neq \emptyset$. Since S is conic (i.e., closed under the natural action of $\mathbb{C}^*$ on the fibers), the fibers of $\pi|_S : S \rightarrow \pi(S)$ have dimension ≥ 1 .

Therefore $\dim \pi(S) < \dim S \leq \dim X$. As a result, there exists $U \subseteq S$ open s.t. $X \setminus \pi(S) \supset U \neq \emptyset$. This implies that $Ch(M|_U) \setminus T_U^*U = \emptyset$, which is what we wanted to show. $\square$

The previous proposition suggests that we may consider holonomic D_X -modules as integrable connections away from a closed proper subset.

Proposition 6.9. *i) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be a short exact sequence of coherent D_X -modules. Then M' & M'' are holonomic $\iff M$ is holonomic.*

ii) Any holonomic D_X -module has finite length.

Proof. i) This follows from 6.3 since $Ch(M) = Ch(M') \cup Ch(M'')$.

ii) This follows from the fact that if $N \subsetneq M$, then $CC(N) < CC(M)$. $\square$

6.4 Duality Functor

Idea for the construction of the dual of a D_X -module: Let M be a left D_X -module. Then $\mathcal{H}om_{D_X}(M, D_X)$ has a natural right D_X -module structure, and $\mathcal{H}om_{D_X}(M, D_X) \otimes_{\mathcal{O}_X} \omega_X^\vee$ is a left D_X -module.

Since $\mathcal{H}om_{D_X}(-, D_X)$ is **not** exact, it is more natural to consider $\mathbb{R}\mathcal{H}om_{D_X}(M, D_X) \otimes_{\mathcal{O}_X} \omega_X^\vee$. The main question is: is this the correct definition for the dual? Let's see an example:

Example 6.10. Let $X = \mathbb{C}$, and $M = D_X/D_X \cdot P$ for some $P \neq 0$. Apply $\mathcal{H}om(-, D_X)$ to

$$0 \rightarrow D_X \xrightarrow{\times P} D_X \rightarrow M \rightarrow 0$$

to get an exact sequence

$$0 \rightarrow \mathcal{H}om_{D_X}(M, D_X) \rightarrow D_X \xrightarrow{P \times} D_X$$

of right D_X -modules. In this example, $\mathcal{E}xt_{D_X}^0(M, D_X) = \mathcal{H}om_{D_X}(M, D_X) = \ker(P \times -) = 0$, while $\mathcal{E}xt_{D_X}^1(M, D_X) \cong D_X/P \cdot D_X$.

So $\mathcal{E}xt_{D_X}^1(M, D_X) \otimes_{\mathcal{O}_X} \omega_X^\vee \cong D_X/D_X \cdot P^*$, where P^* is the formal adjoint of P .

In conclusion, $\mathcal{E}xt^1$ is better suited to be called the "dual" of M than $\mathcal{E}xt^0$. More generally, when $\dim X = n$ and M is holonomic, we will see that only $\mathcal{E}xt_{D_X}^n(M, D_X)$ is nonzero and $\mathcal{E}xt_{D_X}^n(M, D_X) \otimes \omega_X^\vee$ is also holonomic. This motivates the following definition.

Definition 6.11.

$$\begin{aligned} \mathbb{D}M &:= \mathcal{E}xt_{D_X}^n(M, D_X) \otimes_{\mathcal{O}_X} \omega_X^\vee \\ \mathbb{D}M^\bullet &:= \mathbb{R}\mathcal{H}om_{D_X}(M^\bullet, D_X) \otimes_{\mathcal{O}_X} \omega_X^\vee[\dim X] \end{aligned}$$

6.5 Functors and Holonomicity